Tableau Dashboard Build Guide

Product Experience KPIs - Step by Step

Project: **Digital-Product-Analytics-KPI-Dashboard**. You will build one Tableau workbook called **Product_Experience_KPIs**: 11 worksheets and 1 dashboard, using 2 CSV files. No database, no SQL needed for this build - the CSVs are already prepared for Tableau.

HOW TO USE THIS GUIDE

Keep this PDF open next to Tableau and do ONE step at a time, in order. Do not skip steps. After some sections there is a CHECKPOINT box with the exact numbers your screen must show. If your number is different, stop there - the fix is in the Common Problems table on the last page, or re-check the last few steps. If you get stuck, note the step number and ask me with that number.

Files you need (already in the project)

- **session_level.csv** - 290 rows. The main file. Location: dashboards/tableau/data/session_level.csv

- **funnel_long.csv** - 16 rows. Only for the two funnel sheets. Location: dashboards/tableau/data/funnel_long.csv

- The other kpi_*.csv files in that folder are only for checking numbers. Do not build on them.

What the finished dashboard looks like

- Top row: 4 KPI cards - 290 sessions, 167 conversions, 57.59% conversion rate, 28.57% mobile error rate.

- Middle: funnel chart (left) and conversion-by-device chart (right).

- Next row: error impact chart (left) and segment chart (right).

- Bottom: daily trend line across the full width, plus device and segment filters on the side.

Part 1 - Connect the first CSV file (session_level)

Step 1: Open Tableau (Tableau Public or Tableau Desktop). You see the start screen with a blue **Connect** panel on the left.

Step 2: In the Connect panel, under **To a File**, click **Text File**.

Step 3: A file picker opens. Go to the project folder, then **dashboards -> tableau -> data**, select **session_level.csv** and click **Open**.

You should see: the Data Source page. A preview table of the data appears at the bottom.

Step 4: Look at the preview columns. Find **session_date**. Click the small icon above its name and choose **Date** if it is not already a date.

Step 5: Check the icons of the other columns: **session_id, user_id, device_type, user_segment** must show **Abc** (text). **is_converted, has_render_error, reached_page_view, reached_signup_started, reached_form_step_1, event_count, session_duration_sec** must show # (number). If one is wrong, click its icon and change it.

Step 6: At the very bottom of the screen, click the **Sheet 1** tab. You are now on an empty worksheet. The left side is the **Data pane** - it lists all your fields.

Part 2 - Connect the second CSV file (funnel_long)

Step 7: In the top menu, click **Data**, then **New Data Source**.

Step 8: Choose **Text File** again and open **funnel_long.csv** from the same data folder.

You should see: in the Data pane, at the top, there are now TWO data source names. Clicking a name switches the field list below it. This is normal.

Step 9: In the Data pane, right-click the first source name (session_level) and choose **Rename**. Type: **Session Level**. Press Enter.

Step 10: Right-click the second source name (funnel_long), Rename it to: **Funnel Long**.

Step 11: Click **Funnel Long** once. Check that **stage_order** and **sessions** show the # (number) icon. If stage_order shows Abc, go to the Data Source tab at the bottom-left, select Funnel Long at the top of that page, and change its icon to **Number (whole)**. Then come back to Sheet 1.

Part 3 - Create the calculated fields (Session Level)

All fields in this part are created in the **Session Level** source. The method is the same every time: Data pane top-right small down-arrow -> **Create Calculated Field** -> type the name, paste the formula, click OK.

Step 12: Click **Session Level** in the Data pane so it is the active source (its name is highlighted).

Step 13: Click the small down-arrow at the top of the Data pane and choose **Create Calculated Field**.

In the name box type: Total Sessions | Formula box:

```
COUNTD([session_id])
```

Step 14: Click **OK**. A new field **Total Sessions** appears in the Data pane under Measures.

Step 15: Repeat the same method for the next field:

Name: Conversions

```
SUM([is_converted])
```

Step 16: Repeat again:

Name: Conversion Rate

```
SUM([is_converted]) / COUNTD([session_id])
```

Step 17: Now format Conversion Rate as a percentage. Right-click the **Conversion Rate** field in the Data pane -> **Default Properties** -> **Number Format** -> choose **Percentage** -> set decimal places to **2** -> OK.

Step 18: Create the next field:

Name: Drop-offs

```
COUNTD([session_id]) - SUM([is_converted])
```

Step 19: Create the next field:

Name: Error Sessions

```
SUM([has_render_error])
```

Step 20: Create the next field, then format it as Percentage, 2 decimals, exactly like Step 17:

Name: Error Rate

```
SUM([has_render_error]) / COUNTD([session_id])
```

Step 21: Create the next field (it has two lines - type it exactly, including the brackets). Then format it as Percentage, 2 decimals:

Name: Mobile Error Rate

```
SUM(IF [device_type] = "Mobile" THEN [has_render_error] END)
/ SUM(IF [device_type] = "Mobile" THEN 1 END)
```

Step 22: Create the next field. This one is text, not a number:

Name: Error Cohort

```
IF [has_render_error] = 1 THEN "With ERR_MOBILE_RENDER_502" ELSE "Without error"
END
```

Step 23: The Error Cohort field appears under Measures, but it is really a label. Drag it upward in the Data pane and drop it in the **Dimensions** area at the top.

Step 24: Create the last field (used in tooltips):

Name: Avg Session Duration

```
AVG([session_duration_sec])
```

CHECKPOINT A - test your fields before building anything

Drag TOTAL SESSIONS onto the TEXT box in the Marks card (middle-left area). The sheet shows one number: it must be 290. Now also drag CONVERSIONS and CONVERSION RATE onto TEXT. You must see 167 and 57.59%. If Conversion Rate shows 0.58, you missed the formatting in Step 17 - redo it. When the three numbers are right, drag each pill OFF the Marks card so the sheet is empty again.

Part 4 - The 4 KPI cards

Each KPI card is its own tiny worksheet. We make the first one slowly, then copy it three times.

Step 25: Right-click the **Sheet 1** tab at the bottom and choose **Rename**. Type: **KPI - Sessions**

Step 26: Drag **Total Sessions** onto **Text** in the Marks card.

You should see: one big number 290 in the middle of the sheet (it may look small at first).

Step 27: In the Marks card, click **Label**. Set the font size to **28** and click the **Bold (B)** button.

Step 28: In the toolbar at the top, find the dropdown that says **Standard** and change it to **Entire View**. The number now fills the sheet.

Step 29: Hide the sheet title: right-click the title text at the top of the sheet (it says KPI - Sessions) and choose **Hide Title**.

Step 30: Right-click the **KPI - Sessions** tab at the bottom and choose **Duplicate**. A copy appears.

Step 31: Rename the copy to **KPI - Conversions**. On its Marks card, drag the Total Sessions pill OFF, then drag **Conversions** onto Text.

You should see: 167 in the same big style.

Step 32: Duplicate again. Rename to **KPI - Conv Rate**. Replace the field with **Conversion Rate**.

You should see: 57.59%

Step 33: Duplicate again. Rename to **KPI - Mobile Error Rate**. Replace the field with **Mobile Error Rate**.

Step 34: On this sheet ONLY: drag **device_type** to the **Filters** shelf (the empty box above the Marks card). A window pops up: tick **Mobile** only, untick the others, click OK.

You should see: 28.57% - this card is the mobile error rate, which is why it needs the Mobile filter.

CHECKPOINT B

Four tabs at the bottom: KPI - Sessions (290), KPI - Conversions (167), KPI - Conv Rate (57.59%), KPI - Mobile Error Rate (28.57%).

Part 5 - The Funnel sheet (Funnel Long source)

Step 35: Click the **New Worksheet** icon at the bottom (the small page icon with a + next to the tabs). Rename the new sheet: **Funnel**

Step 36: In the Data pane, click **Funnel Long** so it becomes the active source. The field list changes - this is expected. (From now on, always check which source is active before building a sheet.)

Step 37: Drag **device_type** (from Funnel Long) to the **Filters** shelf. Tick **All Devices** only. Click OK.

Step 38: Drag **funnel_stage** to **Rows** (top area).

Step 39: Drag **sessions** to **Columns**. Horizontal bars appear.

Step 40: Fix the stage order. Click the small down-arrow on the **funnel_stage** pill in Rows and choose **Sort**. In the window: choose **Sort By: Field**, select field **stage_order**, order **Ascending**. Close the window.

You should see: top to bottom: Page View, Signup Started, Form Step 1 Completed, Onboarding Completed. If the order is alphabetical, you missed this step.

Step 41: Drag **sessions** (again, from the Data pane) onto **Label** in the Marks card. The numbers 290, 241, 201, 167 appear on the bars.

Step 42: Drag **cumulative_retention_pct** onto **Label** as well. Two numbers now sit on each bar.

Step 43: Format the retention label. Click the **cumulative_retention_pct** pill inside the Marks card, choose **Format**. In the format panel, find **Numbers**, choose **Custom**, and in the format box type exactly: 0.00"%"

CAREFUL - the percent trap in this sheet

Do NOT use the normal Percentage format for cumulative_retention_pct. The value in the file is already 83.10, meaning 83.10 percent. Percentage format multiplies by 100 and would show 8310%. The custom format 0.00"%" only adds the % sign. (This is the opposite of Conversion Rate in Step 17, which holds 0.5759 and DOES need Percentage format.)

Step 44: Also drag **stage_retention_pct** onto **Tooltip** in the Marks card, and format it the same custom way (0.00"%") if it shows wrongly when you hover.

Step 45: Drag **funnel_stage** onto **Color** in the Marks card. Each stage gets its own colour.

CHECKPOINT C

Bars read 290, 241, 201, 167 with cumulative labels 100.00%, 83.10%, 69.31%, 57.59%. Compare with the file kpi_funnel_stage.csv - they must match.

Part 6 - Device Funnel sheet (still Funnel Long)

Step 46: Right-click the **Funnel** tab and choose **Duplicate**. Rename the copy: **Device Funnel**

Step 47: On the Filters shelf, open the **device_type** filter. Untick All Devices. Tick **Desktop, Mobile, Tablet**. Click OK.

Step 48: Drag **device_type** to **Columns** and drop it on the LEFT side of the SUM(sessions) pill. The funnel splits into three side-by-side funnels, one per device.

CHECKPOINT D

Desktop: 142, 134, 114, 101. Mobile: 98, 61, 50, 36. Tablet: 50, 46, 37, 30.

Part 7 - Device Performance sheet (back to Session Level)

Step 49: Create a **New Worksheet**. Rename it: **Device Performance**

Step 50: FIRST click **Session Level** in the Data pane to switch the source back. Most mistakes in this build come from building a sheet on the wrong source.

Step 51: Drag **device_type** to **Rows**.

Step 52: Drag **Conversion Rate** to **Columns**. Three bars appear.

Step 53: Sort the bars. Right-click the **device_type** pill in Rows -> **Sort** -> Sort By: **Field** -> field **Conversion Rate** -> **Descending** -> Close. Order becomes Desktop, Tablet, Mobile.

Step 54: Drag **Conversion Rate** onto **Label**. The percentages appear on the bars.

Step 55: Colour the bars. Drag **device_type** onto **Color**. Then click **Color** -> **Edit Colors**. Click Desktop and pick a green, click Tablet and pick an orange, click Mobile and pick a red. (For the exact colours, use More Colors and type the hex: Desktop 55A868, Tablet DD8452, Mobile C44E52.) Click OK.

CHECKPOINT E

Desktop 71.13%, Tablet 60.00%, Mobile 36.73%.

Optional but useful: the Device Table sheet

Step 56: Duplicate the Device Performance sheet. Rename the copy: **Device Table**

Step 57: On the copy, drag the Conversion Rate pill OFF Columns (sheet becomes empty). Then drag these fields one by one onto **Text**: Total Sessions, Conversions, Conversion Rate, Drop-offs, Error Sessions, Error Rate. Tableau builds a table automatically (you will see Measure Names / Measure Values appear - that is normal).

Device	Sessions	Conv.	Rate	Drop-offs	Errors	Err rate
Desktop	142	101	71.13%	41	0	0.00%
Mobile	98	36	36.73%	62	28	28.57%
Tablet	50	30	60.00%	20	0	0.00%

Your Device Table must match this table exactly. If one number differs, that sheet has a leftover filter - open its Filters shelf and clear it.

Part 8 - Error Impact sheet (Session Level)

Step 58: Create a **New Worksheet** (Session Level source). Rename it: **Error Impact**

Step 59: Drag **device_type** to **Filters**. Tick **Mobile** only. Click OK. (This whole sheet is mobile-only.)

Step 60: Drag **Error Cohort** to **Columns**.

Step 61: Drag **Total Sessions** to **Rows**. Two bars appear.

Step 62: Drag **Conversion Rate** onto **Label**.

Step 63: Drag **Error Cohort** onto **Color**, then Edit Colors: **With ERR_MOBILE_RENDER_502** = red (C44E52), **Without error** = green (55A868).

CHECKPOINT F

With ERR_MOBILE_RENDER_502: 28 sessions, label 0.00%. Without error: 70 sessions, label 51.43%. This is the most important chart in the dashboard - it proves the error kills conversion.

Part 9 - Segments sheet (Session Level)

Step 64: Create a **New Worksheet**. Rename it: **Segments**

Step 65: Drag **user_segment** to **Rows**.

Step 66: Drag **Conversion Rate** to **Columns**.

Step 67: Sort descending, the same way as Step 53 (Sort By Field: Conversion Rate, Descending).

Step 68: Drag **Conversion Rate** onto **Label**. Also drag **Error Sessions** and **Total Sessions** onto **Tooltip** (they appear when the mouse hovers a bar).

CHECKPOINT G

Premium 64.86% (74 sessions), Enterprise 60.00% (50), Free 53.61% (166).

Part 10 - Daily Trend sheet (Session Level)

Step 69: Create a **New Worksheet**. Rename it: **Daily Trend**

Step 70: Drag **session_date** to **Columns**.

Step 71: Click the down-arrow on the session_date pill and choose **Day** from the SECOND (lower) group of date options. The pill turns GREEN - that means continuous dates, which is what a trend line needs. If the pill stays blue, you picked the wrong Day - undo and pick the lower one.

Step 72: From the bottom of the Data pane, drag **Measure Values** to **Rows**. Many lines appear - do not worry, we remove the extras next.

Step 73: A **Measure Values** card appears under the Marks card, listing fields. Drag OFF every field in that list except **Conversion Rate** and **Mobile Error Rate**.

Step 74: From the bottom of the Data pane, drag **Measure Names** onto **Color** in the Marks card. Now exactly two lines remain, in two colours, with a legend.

CHECKPOINT H

The lines run from 1 Sep 2026 to 14 Sep 2026 (14 days). Daily values are noisy - that is expected with about 21 sessions a day.

Part 11 - Assemble the dashboard

Step 75: Click the **New Dashboard** icon at the bottom (the icon that looks like a small grid/window, next to New Worksheet). Rename the tab: **Product Experience KPIs**

Step 76: On the left, in the Dashboard pane under **Size**, choose **Fixed size** and set Width **1280** and Height **850** if your version shows width/height boxes. If it shows presets instead, pick the largest desktop preset.

Step 77: On the left you see the list of your Sheets. Drag **KPI - Sessions** onto the empty dashboard (drop it where it says Drop sheets here).

Step 78: Drag **KPI - Conversions** and drop it on the RIGHT EDGE of the first card, so it lands beside it, not below. Do the same for **KPI - Conv Rate** and **KPI - Mobile Error Rate**. You now have one row of 4 KPI cards. (Watch the grey shadow while dragging - it shows where the sheet will land.)

Step 79: Drag **Funnel** below the KPI row, on the left half. Drag **Device Performance** on the right half, next to it.

Step 80: If you made the Device Table sheet, drag it under Device Performance. (Optional.)

Step 81: Next row: drag **Error Impact** on the left and **Segments** on the right.

Step 82: Drag **Daily Trend** at the bottom, across the full width.

Step 83: For each tile: if a sheet looks squeezed, click the tile once, then use its small down-arrow (top-right of the tile) -> **Fit** -> **Entire View**.

Step 84: Add the dashboard title. Double-click the title area at the top of the dashboard and type: **Digital Product Experience - KPI Dashboard**. If there is no title area, drag a **Text** object (left side, Objects) to the top and type it there, bold, size 16.

Step 85: Add the insight text. Drag a **Text** object (Objects, left side) under the title or into a free corner, and type this text:

TEXT TO TYPE IN THE INSIGHT BOX

Headline: Mobile converts at 36.73% vs Desktop 71.13%. Cause: ERR_MOBILE_RENDER_502, a rendering error on /onboarding-1 that hits 28.57% of mobile sessions - every affected session drops out and none convert. Fix value: about +14 conversions per 98 mobile sessions (mobile to about 51.4%); upper bound 65.31% if every error session converted.

Step 86: Add the device filter. Click the **Device Performance** tile once to select it. Click its small down-arrow (top-right of the tile) -> **Filters** -> **device_type**. A filter box appears on the right side of the dashboard.

Step 87: Click the down-arrow of that new filter box -> **Apply to Worksheets** -> **Selected Worksheets** -> tick ONLY: Device Performance, Device Table (if used), Segments, Daily Trend -> OK.

Step 88: Add the segment filter the same way: select the **Segments** tile -> its down-arrow -> Filters -> **user_segment** -> then Apply to Worksheets -> Selected Worksheets -> tick the same Session Level sheets as Step 87.

CAREFUL - filters and the funnel sheets

Do NOT apply these dashboard filters to the Funnel and Device Funnel sheets. They use the other data source (Funnel Long) and already have their own device filter from Steps 37 and 47. Applying the dashboard filter to them can make them go blank.

Step 89: Play with the two filters once to test: choose Mobile in the device filter. The device, segment and trend charts should change; the 4 KPI cards and the funnel should NOT change. Then set the filter back to All.

Part 12 - Save and publish

Step 90: Save the file. Menu **File -> Save As**. Name it exactly: **Product_Experience_KPIs**. Save it inside the project's **dashboards** folder.

Step 91: If Tableau asks for a format, choose **Packaged Workbook (.twbx)** - this packs the CSV data inside the file, so the workbook still opens on another computer.

Step 92: Tableau Public users: instead of Save As, use **Server -> Publish Workbook** (or Save to Tableau Public), sign in, and publish with the same name.

Step 93: After publishing, the workbook opens in your browser. Copy its URL - this is your Tableau Public link. Paste it into dashboards/README.md in the project so it is saved with the project.

Final check - your dashboard must show these numbers

Place	Must show
KPI cards	290 | 167 | 57.59% | 28.57%
Funnel	290 - 241 - 201 - 167 (cumulative 100 / 83.10 / 69.31 / 57.59%)
Device rates	Desktop 71.13% | Tablet 60.00% | Mobile 36.73%
Error Impact	28 sessions at 0.00% vs 70 sessions at 51.43%
Segments	Premium 64.86% | Enterprise 60.00% | Free 53.61%
Daily Trend	14 days, 1 Sep to 14 Sep 2026

Common problems and fixes

Problem	Fix
Conversion Rate shows 5759%	The field needs Percentage format - redo Step 17.
Funnel labels show 8310%	You used Percentage format on a retention field. Use the custom format 0.00"%" - Step 43.
Funnel stages in A-B-C order	Sort funnel_stage by the stage_order field - Step 40.
Funnel shows 1000 sessions	You built it on the wrong file. Funnel sheets use funnel_long.csv data only - Steps 36-37.
A sheet shows wrong numbers	Check which data source is active (Step 50) and clear leftover filters on that sheet.
KPI cards change with filters	Filters were applied to all worksheets. Redo Steps 87-88 with Selected Worksheets only.
Trend shows months, not days	The date pill is blue (discrete). Pick Day from the lower group so it turns green - Step 71.
A dashboard tile is blank	A filter from another data source was applied to it - see the Careful box after Step 88.

If you are stuck at any point, tell me the **step number** where you stopped and what your screen shows - I will walk you through that one step.